Anatomy & Kinesiology: Wrist & Hand

BONES AND JOINTS

Distal radius and ulna:
- **Ulnar tilt 25°:** Reason why ulnar deviation > radial deviation.
- **Palmar tilt 10°:** Reason why wrist flexion > wrist extension.

Carpal bones:

- **Proximal row:** Scaphoid, Lunate, Triquetrum, Pisiform
- **Distal row:** Trapezium, Trapezoid, Capitate, Hamate
  Other terms:
  - Scaphoid = navicular
  - Trapezium = greater multangular
  - Trapezoid = lesser multangular
  - Triquetrum = triangular
  - Capitate = Os magnum
- **Lister's tubercle:** dorsal tubercle separating tendons of EPL and ECRB.

WRIST LIGAMENTS

Extrinsic ligaments:

- Dorsal radiocarpal
- Radial collateral
- Palmar radiocarpal
- Radioscapholunate
- Radiolunate
- Radioscaphocapitate
- **Triangular Fibrocartilage Complex (TFCC):**
  - **Primary stabilizer of distal RU joint**
  - Reinforces ulnar side of wrist
  - Helps transfer compressive forces from hand to forearm
- RU joint capsular ligament
- Palmar ulnocarpal (ulnotriquetral, ulnolunate)
- Ulnar collateral ligament
- Meniscus homologue

Intrinsic ligaments:

- **Short (distal row):** dorsal, palmar, interosseous
- **Intermediate:** Lunotriquetral, Scapholunate, Scaphotrapezial
- **Long:**
  - **Palmar intercarpal (inverted V):**
    - Lateral leg capitate to scaphoid
    - Medial leg capitate to triquetrum
  - **Dorsal intercarpal:** trapezium, scaphoid, lunate, triquetrum

WRIST KINEMATICS

- **RC motion:** greater in flexion (50°), MC motion (35° flexion)
- **MC motion:** greater in extension (50°), RC (35° extension)
- **Radial deviation:**
  - Roll: lateral
  - Glide: medial
  - Ligaments taut: ulnar collateral, palmar radiocarpal (RC), palmar intercarpal medial leg (MC)
- **Ulnar deviation:**
  - Roll: medial
  - Glide: lateral
  - Ligaments taut: radial collateral, palmar ulnocarpal (RC), palmar intercarpal lateral leg (MC)

METACARPALS (5)

- 2nd MCP abduction: roll + glide lateral
- 3rd MCP joint = axis
- 4th/5th MCP abduction: roll + glide medial
- **Deep transverse ligament:** prevents splay deformity

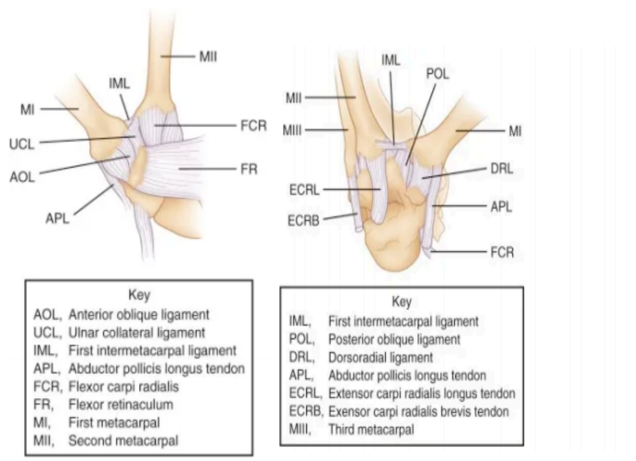
THUMB CARPOMETACARPAL (CMC) JOINT

- The ligaments of the thumb CMC joint are located on all sides, contributing to both stability and mobility.
- **Anterior oblique ligament (beak ligament):** Connects the palmar tubercle of the first metacarpal base to the trapezium.
  - Primary stabilizing ligament of the joint and becomes taut during abduction, extension, and opposition.
- **Ulnar collateral ligament:** Found on the volar and medial side, extending from the transverse carpal ligament to the

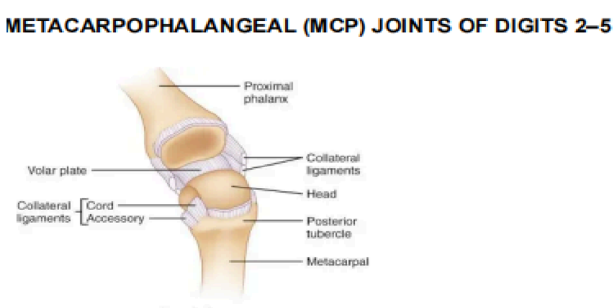
palmar-medial base of the first metacarpal.

- Provides medical stability.

- **Posterior oblique ligament:** Lies dorsally and connects the trapezium to the medial base of the first metacarpal.
- **Dorsoradial ligament:** Extends from the dorsolateral trapezium to the dorsal base of the first metacarpal; an important dorsal stabilizer.
- **Intermetacarpal ligament:** Connects the bases of the first and second metacarpals.
- **Joint capsule:** Complete but relatively loose, allowing axial rotation for opposition.



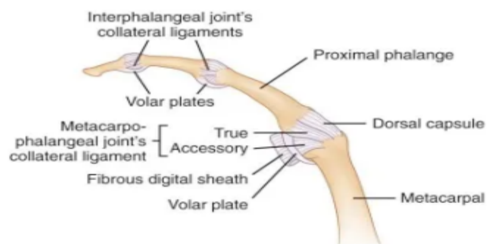
METACARPOPHALANGEAL (MCP) JOINTS OF DIGITS 2-5



The MCP joints are formed by the convex heads of the metacarpals and the concave bases of the proximal phalanges.

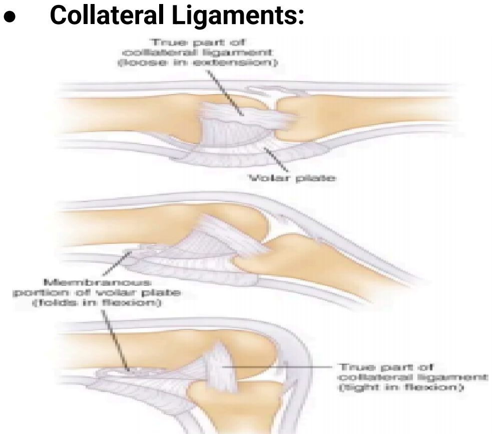
- **Metacarpal head: Biconvex and cam-shaped**, extending further volarly and being wider on the volar side.
    - The base of the proximal phalanx is **biconcave and smaller.**
    - Allows movement in multiple planes but actual motion is limited by soft tissues.
    - The MCP joints allow **flexion, extension, abduction, and adduction.** *(Rotation: possible passively; abd. and add.: occur relative to the middle finger)*

- **Joint Capsule and Volar Plate:**
  - The MCP joint capsule **encloses the joint** and is specialized on the volar side by the volar plate, **a fibrocartilaginous structure.**



- The volar plate **prevents hyperextension and provides attachment for the A1 pulley of the flexor tendon sheath**, helping maintain proper positioning of the flexor tendons.
- Distally, it is **thick and firmly attached** to the proximal phalanx.
- Proximally, it is **thinner, flexible, and folds** during flexion.

- **Collateral Ligaments:**

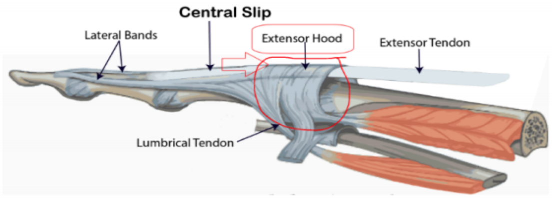


- **True (proper) collateral ligament:** Dorsal metacarpal tubercle to volar proximal phalanx.
    - Loose in extension

(**allows abduction/adduction), tight in flexion (cam shape limits side-to-side motion**).
    - **Accessory collateral ligament:** Attaches to the volar plate; stabilizes the joint and flexor tendons. (loosens slightly during flexion)

EXTENSOR HOOD

- **Fibrous structure surrounding the joint**



- Reinforces MCP joints.
- Includes **sagittal bands** to **keep extensor tendons centered.**
- Transmits force from extensor tendons **to extend the proximal phalanx.**
- **Moves proximally and distally during motion.**

THUMB MCP JOINT

- The volar plate contains two **sesamoid bones** (groove for FPL).
    - **Ulnar side hood:** Formed by adductor pollicis (stronger).
    - **Radial side hood:** Formed by APB and FPB (Abductor and Flexor Pollicis Brevis).
- More stable but vulnerable to **ulnar collateral ligament injury.**

Arthrokinematics:
- Abduction: roll anterior, glide posterior
- Flexion: roll + glide laterally
- Opposition
    - **Phase 1**: roll anterior, glide posterior
    - **Phase 2**: roll + glide medially

PROXIMAL INTERPHALANGEAL (PIP) JOINT

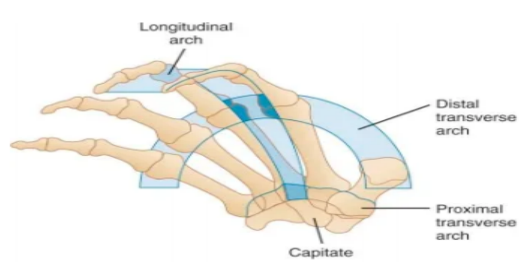
- The PIP joint is formed by the **head of the proximal phalanx** and the **base of the middle phalanx.**
- The **head is cylindrical**, and the **base is concave**, with a sagittal groove and ridge that enhance stability. This design **restricts movement to a single plane.**
- The PIP joint allows **only flexion and extension.**

PIP JOINT SUPPORTING STRUCTURES:
- Cylindrical head (proximal phalanx) and concave base (middle phalanx).
- **Volar plate:** prevents hyperextension; includes **check-rein ligaments** (tighten during extension to limit excessive motion).
- Attachment for **A3 pulley.**
    - **True collateral ligament:** Remains tight throughout the range of motion, **providing constant stability.**
    - Accessory ligament: supports volar plate.
- Laterally, the lateral bands and retinacular ligaments provide support.
- Both the flexor digitorum superficialis and profundus tendons pass volarly across the joint.

DISTAL INTERPHALANGEAL (DIP) JOINT

- Structural similar to PIP.
- **Only allows flexion and extension** due to its hinge-like structure.
- The volar plate prevents hyperextension and provides attachment for **A5 pulley**.
- Only **Flexor Digitorium Profundus** crosses volarly. (dorsally the extensor mechanism is less complex, primarily involving the terminal tendon).

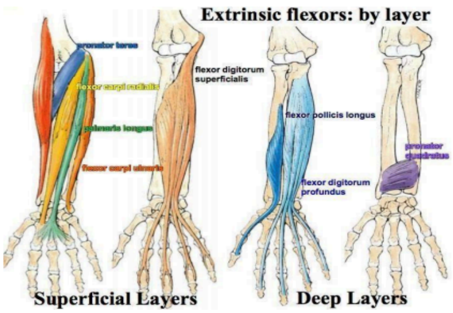
ARCHES OF THE HAND

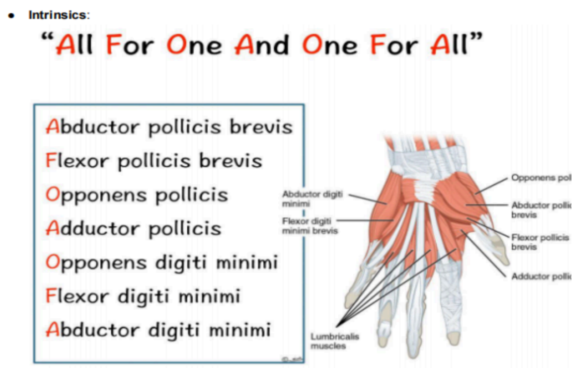


- **Proximal transverse arch:** Keystone = capitate
- **Distal transverse arch:** Keystone = middle

finger
- **Longitudinal arch:** Keystone = between 2nd/3rd finger

MUSCLES OF WRIST & HAND





Extrinsics:

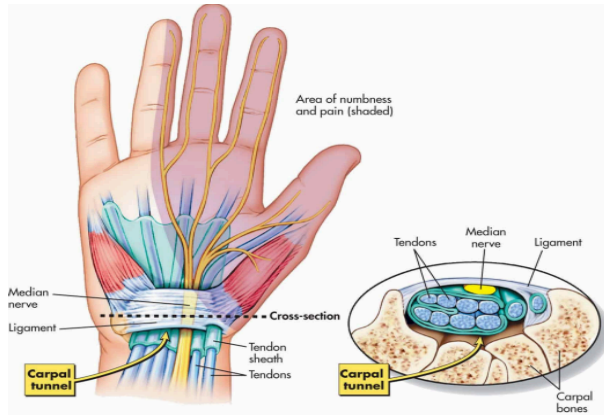
- **Anterior compartment (superficial):** FCR, PL, FCU (ulnar nerve)
- **Intermediate:** FDS
- **Deep:** FPL, FDP (flex DIP, dually innervated; only muscle active in unresisted closure)
- **Lateral compartment (radial nerve):** Brachioradialis, ECRL
- **Posterior superficial:** ECRB, ED, EDM, ECU (wrist extensors increase grip strength)
- **Posterior deep:** APL, EPB, EPL, EIP

Intrinsics:

- **"All For One And One For All":** APB, FPB, OP, AP, ODM, FDM, ADM
- **Lumbricals:** silent in MCP flexion if IP joints flexed.
- **Interossei:** stabilize extensor hood, pinch, grasp.

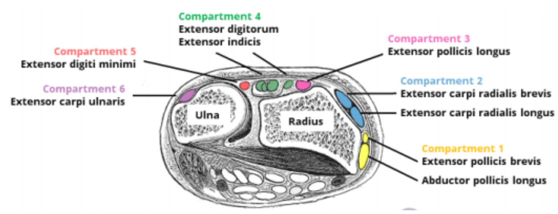
SPECIAL TOPICS

Carpal Tunnel:



- **Roof:** Transverse carpal ligament (pisiform/hamate to scaphoid/trapezium)
- **Contents:** 4 FDP, 4 FDS, 1 FPL, and Median Nerve.

Extensor Tunnel (6 Compartments):

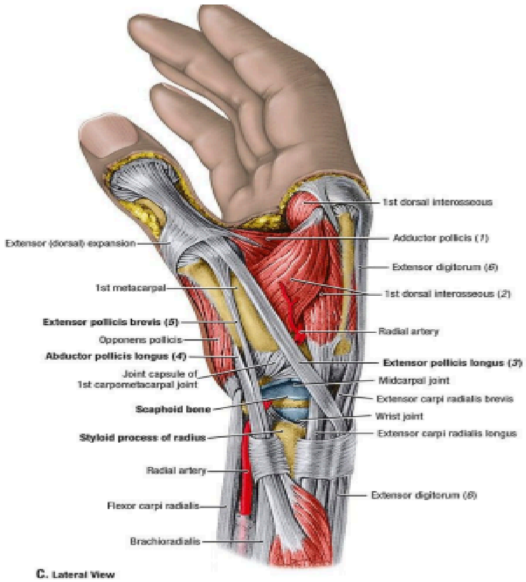


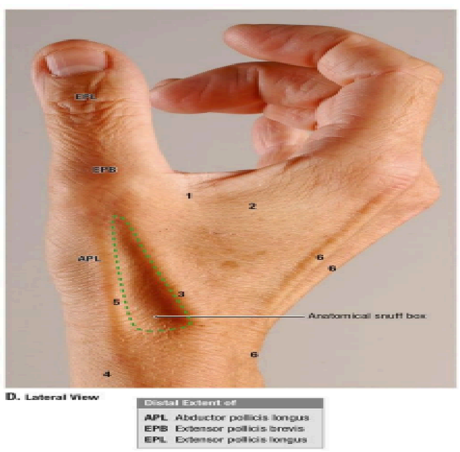
- Attachments - pisiform and hook of hamate to distal radius.

1. APL, EPB
2. ECRL, ECRB
3. EPL
4. EDC, EIP
5. EDM
6. ECU

Anatomic Snuffbox:

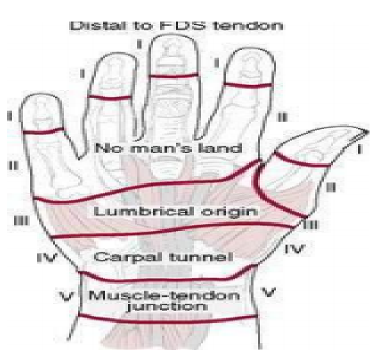
Anatomy & Kinesiology: Wrist & Hand





- **Borders:** APL/EPB (radial), EPL (ulnar)
- **Floor:** Scaphoid/Trapezium
- **Content:** Radial artery

No Man's Land:

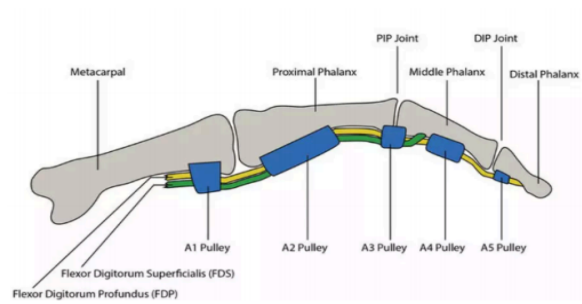


- Between distal palmar crease and PIP crease.

- Tendon repair is contraindicated due to scarring.

- **Extensor assembly**:
  - Tendinous - extrinsic extensors + intrinsic muscles
  - Retinacular - fascia + ligaments
  - Components:
    - **Central band** — transmits ED force across IP
    - **Lateral bands** — transmits force across PIP/DIP
    - **Dorsal hood** (transverse & oblique fibers)
    - **Oblique retinacular ligament** — coordinates PIP & DIP motion

Flexor Pulley System



- Series of **fibrous bands that hold the flexor tendons** (flexor digitorum superficialis **and flexor digitorum profundus close to the bones of the fingers.**
- **Prevent bowstringing during finger flexion** and to ensure efficient force transmission from muscle to bone.

- **GENERAL FUNCTION**
  - When the flexor tendons contract, they tend to pull away from the bone. **The pulleys act like restraints that keep the tendons close to the phalanges**, allowing smooth and efficient flexion.
  - **Without pulleys, finger flexion would be weak and mechanically inefficient**.

Anatomy & Kinesiology: Wrist & Hand

- **CLASSIFICATION OF PULLEYS**
  - There are two types of pulleys in each finger:

1. **Annular pulleys (A pulleys)**
2. **Cruciform pulleys (C pulleys)**

- **Annular Pulleys (A1-A5):**
  - **A1:** Level of MCP; **common site of trigger finger.**
  - **A2:** Proximal phalanx; **most important to prevent bowstringing**; maintains optimal tendon biomechanics.
  - **A3:** PIP joint.
  - **A4:** Middle phalanx; works with A2 for tendon efficiency.
  - **A5:** DIP joint.

- **Cruciform Pulleys (C1-C3): Thinner and more flexible**; located between the annular pulleys
  - Allow the tendon sheath to collapse and expand during finger movement.
  - **Prevent stiffness or constriction of tendon sheath during flexion.**

BIOMECHANICAL IMPORTANCE

- A2 and A4 pulleys - most critical for normal finger function.
  - If these pulleys are damaged, bowstringing occurs, where the tendon pulls away from the bone, resulting in reduced grip strength and inefficient motion.

CLINICAL CORRELATIONS

- **Trigger finger -** Caused by thickening or inflammation of the A1 pulley.
  - Leads to **catching or locking of the finger** during movement.
- **Pulley rupture - Common in rock climbers** due to excessive load.
  - Most commonly affects A2 pulley.
  - Leads to visible bowstringing and weakness.
- **Flexor tendon injury -** Pulley integrity must be preserved during repair to maintain function.

POWER AND PRECISION GRIPS

Grip	Description	Finger Function	Thumb Function	Example
Lateral pinch (key)	Object between index finger & thumb	Index abducted, partially flexed MCP/IP; FDP, FDS, 1st dorsal interossei active	Thumb abducted, IP flexed; FPL, FPB, adductor pollicis	Holding keys, grasping paper
Three-prong chuck	Pads of 2nd, 3rd digits & thumb	MCP/PIP flexion; FDS (if DIP extended), FDP (if flexed)	Thumb in opposition, flexed MCP/IP, adduction at CMC; adductor pollicis stabilizes	Holding coins, writing
Tip-to-tip grip	Tip prehension between thumb & finger	Full flexion of fingers; FDP primary mover	Thumb in opposition, flexed MCP/IP; FPL important	Picking up small items
Hook grip	Used for carrying objects	FDS & FDP active, IP flexion	Thumb not involved	Carrying bags, weights
Cylinder grip	Holding cylindrical objects	Flexion of MCP/IP; interossei counterbalance radial ligaments	Thumb flexed/adducted, opposing fingers	Holding bottles, phone
Spherical grip	Holding round objects	MCP abduction, partial flexion; extrinsic flexors, ED co-contract	Thumb opposition, MCP/IP flexion, thenar muscles active	Holding a ball or apple
Fist grip	Thumb wraps around object	Flexion of all finger joints	Thumb adducted or opposed to fingers	Gripping small tools, bat

Power Grip

Figure 7-39 Types of power grips.

Precision Grip

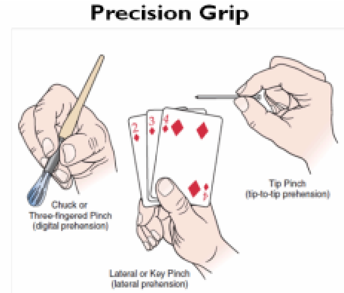


Figure 7-40 Types of precision grips or pinches.